

Notes — Biology Chapter 7: Metabolism

Quick reference. Clean. No fluff. For the "why" behind each idea, see [explanations.md](#).

Topic 1 — Metabolism and Its Types

Definitions

Term	Definition
Metabolism	The sum of all chemical reactions occurring within each cell of a living organism for the maintenance and sustainability of life
Catabolism	Biochemical reactions that break down complex molecules into simpler ones; energy is released
Anabolism	Biochemical reactions that build complex molecules from simpler ones; energy is consumed
Biochemical reactions	Chemical reactions taking place inside living organisms

Catabolism vs Anabolism

Feature	Catabolism	Anabolism
Direction	Complex → Simple	Simple → Complex
Energy	Released	Consumed
Example 1	Digestion (large food → small molecules)	Photosynthesis ($\text{CO}_2 + \text{H}_2\text{O} \rightarrow \text{glucose}$)
Example 2	Cellular respiration ($\text{glucose} \rightarrow \text{CO}_2 + \text{H}_2\text{O}$)	Glycogenesis (glucose → glycogen)

Key point: Energy released in catabolism is utilized in anabolism.

Topic 2 — Enzymes and Their Characteristics

Definition

Enzymes are biologically active globular proteins made by living cells, which tremendously speed up biochemical reactions.

- Enzymes = biological catalysts
- Substrates = molecules enzymes act on
- Products = substrates modified by enzymes
- Study of enzymes = **enzymology**

Types of Enzymes

Type	Location	Example
Intracellular	Work inside cells	Enzymes of glycolysis (cytoplasm)
Extracellular	Released to work outside	Pepsin (stomach cavity)

8 Characteristics of Enzymes

#	Characteristic	Key detail
1	Globular proteins	Chains of amino acids folded into 3D globular shape; some are RNA (ribozymes)
2	Increase reaction rate	Speed up reactions millions of times; do NOT start reactions
3	Required in small quantity	Regain original shape after reaction; used repeatedly
4	Specific	Each enzyme catalyses only one type of reaction (one substrate)
5	Require cofactor	Many need non-protein component for proper function
6	Regulated	Activity can be increased or decreased by cell according to requirements
7	Make metabolic pathways	Work in sequence; metabolic pathway = series of connected reactions
8	Lower activation energy	Reduce energy needed to start a reaction so it happens at body temperature

Cofactors

Type	Description	Examples
Activator	Detachable inorganic ions	Zn ²⁺ , Fe ²⁺ , Cu ²⁺ , chloride ions
Prosthetic group	Organic molecule tightly/permanently attached to enzyme	FAD (Flavin Adenine Dinucleotide), Haem group
Coenzyme	Detachable organic molecule	NAD, Coenzyme A, Vitamin A, Vitamin C

Topic 3 — Mechanism of Enzyme Action

Key Terms

Term	Definition
Active site	Small, charged-bearing cavity on enzyme surface where substrate binds
ES Complex	Enzyme-Substrate Complex; formed when substrate binds to active site
EP Complex	Enzyme-Product Complex; ES complex after catalysis occurs

Steps of Enzyme Action

Enzyme + Substrate → ES Complex → EP Complex → Products + Enzyme (unchanged)

1. Substrate binds to active site → forms ES Complex
2. Enzyme catalyses conversion of substrate to product → forms EP Complex
3. EP Complex breaks apart → products released, enzyme released unchanged
4. Enzyme is available again for the same reaction

Activation energy: The minimum extra energy needed to start a reaction. Enzymes lower this so reactions occur at body temperature.

Topic 4 — Factors Affecting Enzyme Activity

4.1 Temperature

Condition	Effect
Low temperature	Enzyme inactive (low activation energy available); activity resumes at normal temperature
Increasing temperature	Rate of reaction increases
Optimum temperature	Maximum activity (37°C for human enzymes)
High temperature (>50°C)	Enzyme denatured — active site damaged, activity lost permanently

4.2 pH

Condition	Effect
Below optimum pH	Rate of reaction decreases
Optimum pH	Maximum activity
Above optimum pH	Rate of reaction decreases
Extreme pH change	Enzyme denatured

Enzyme	Location	Optimum pH
Pepsin	Stomach	pH 2.0 (acidic)
Trypsin	Small intestine	pH 8.0 (alkaline)

4.3 Substrate Concentration

Condition	Effect
Low substrate, many enzymes	Rate proportional to substrate concentration
Increasing substrate	Rate of reaction increases
All active sites occupied	Rate plateaus — saturation of active site
Further increase in substrate	No further effect

Inhibitors and Activators

Term	Definition
Inhibitors	Molecules that stop enzymatic reactions
Activators	Molecules that increase enzymatic reaction rate

Topic 5 — Enzyme Inhibition

Definition

Enzyme inhibition: phenomenon where enzymes cannot perform their role due to interference of chemicals at the reaction site.

Inhibitor: a chemical substance that reacts with enzyme in place of substrate but does not convert to products, thus inhibiting enzyme action.

Competitive vs Non-Competitive Inhibition

Feature	Competitive	Non-Competitive
Resembles substrate?	Yes — similar shape	No — different shape
Binds at?	Active site	Allosteric site (different from active site)
Reversible?	Yes — if substrate concentration increased	No — damages active site permanently

Mechanism	Blocks substrate from binding	Distorts enzyme shape → active site damaged
Result	No product formed	No product formed

Allosteric site: the binding point for non-competitive inhibitors, located away from the active site.

Topic 6 — ATP: The Energy Currency of the Cell

Definition

ATP (Adenosine Triphosphate) is the universal energy currency of cells — an energy-rich compound that stores and releases energy on demand.

Structure of ATP

Component	Details
Adenine	Double-ringed nitrogen base
Ribose	Five-carbon sugar
Phosphate groups	Three phosphate groups (PO ₄) linked in a chain = triphosphate group

Assembly:

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Adenine + Ribose → Adenosine
Adenosine + 1 PO4 → AMP (Adenosine Monophosphate)
AMP + 1 PO4 → ADP (Adenosine Diphosphate)
ADP + 1 PO4 → ATP (Adenosine Triphosphate)

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High-energy bonds: The covalent bonds between phosphate groups store energy (shown as wavy lines ~P).

ATP-ADP Cycle

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ATP + H2O → ADP + Pi + ENERGY (hydrolysis – energy released for cell work)
ADP + Pi + ENERGY → ATP (condensation – energy stored from catabolism/light)

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Science Titbit: Human cells hydrolyse 100–150 moles of ATP per day. One glucose molecule in aerobic respiration produces a net 36 ATP molecules.

Topic 7 — Photosynthesis

Overall Equation

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6CO2 + 12H2O →(Light, Chlorophyll)→ C6H12O6 + 6H2O + 6O2

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Two Phases

Phase	Location	Needs light?	What happens

Light-dependent reactions	Thylakoid membranes (grana)	Yes	Water split (photolysis), ATP & NADPH produced, O ₂ released
Light-independent reactions (Calvin cycle / dark reactions)	Stroma of chloroplast	No	CO ₂ fixed using ATP & NADPH to make glucose (G3P)

Light-Dependent Reactions — Steps

1. Photosystem II absorbs light → electrons emitted from chlorophyll a
2. Water undergoes **photolysis**: $\text{H}_2\text{O} \rightarrow \text{O}_2 + 2\text{H}^+ + 2\text{e}^-$ (replaces lost electrons)
3. Oxygen released through stomata
4. Electrons pass through **Electron Transport Chain (ETC)** → energy used for ATP synthesis
5. Photosystem I absorbs light → electrons reduce NADP⁺ to **NADPH**

Light-Independent Reactions (Calvin Cycle) — Steps

1. CO₂ combines with **RuBP** (5-carbon) → unstable 6-carbon intermediate
2. 6-carbon molecule splits → two **3-PGA** (3-carbon) molecules
3. 3-PGA reduced by NADPH + ATP → **G3P** (3-carbon sugar)
4. Some G3P → glucose; some G3P regenerates RuBP

Limiting Factors of Photosynthesis

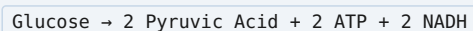
Factor	Effect
Light intensity	More light → faster photosynthesis
CO ₂ concentration	More CO ₂ → faster (up to a point)
Temperature	Optimum = 25°C; too high or too low reduces rate

Topic 8 — Respiration

Overall Equation (Aerobic)



Glycolysis (common to both aerobic and anaerobic)



- Location: **cytoplasm**
- Occurs in: both aerobic and anaerobic respiration

Anaerobic Respiration

Respiration without oxygen. Incomplete oxidation of glucose.

Type	Reaction	Where
Alcoholic fermentation	Pyruvic acid → Ethyl alcohol + CO ₂	Yeast, some bacteria
Lactic acid fermentation	Pyruvic acid → Lactic acid	Muscle cells during vigorous exercise; bacteria

Aerobic Respiration — 4 Stages

Stage	Location	Inputs	Outputs
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1. Glycolysis	Cytoplasm	Glucose	2 Pyruvic acid, 2 ATP, 2 NADH
2. Formation of Acetyl CoA	Mitochondria	2 Pyruvic acid	2 Acetyl CoA, 2 CO ₂ , 2 NADH
3. Krebs Cycle	Mitochondrial matrix	2 Acetyl CoA	4CO ₂ , 2 ATP, 6 NADH, 2 FADH ₂
4. Electron Transport Chain	Inner mitochondrial membrane	NADH, FADH ₂ , O ₂	32 ATP, H ₂ O

Net ATP from aerobic respiration: 36 ATP per glucose molecule

Aerobic vs Anaerobic Comparison

Feature	Aerobic	Anaerobic
Oxygen needed?	Yes	No
Glucose fully broken down?	Yes	No
Final products	CO ₂ + H ₂ O	CO ₂ + alcohol OR lactic acid
ATP yield	36 ATP	2 ATP
Location	Cytoplasm + mitochondria	Cytoplasm only

Must-Remember Key Points

- Metabolism = catabolism + anabolism
- Enzymes = biological catalysts; they lower activation energy but are not consumed
- Active site = where substrate binds; specific to ONE substrate type
- Optimum temperature for human enzymes = 37°C; denaturation occurs at high temperatures
- Cofactors: activators (inorganic ions), prosthetic groups (permanent organic), coenzymes (detachable organic)
- Competitive inhibition: reversible, blocks active site; non-competitive: irreversible, blocks allosteric site
- ATP = Adenosine + 3 phosphate groups; energy stored in high-energy phosphate bonds
- Photosynthesis equation: $6\text{CO}_2 + 12\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{H}_2\text{O} + 6\text{O}_2$
- Aerobic respiration equation: $\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 \rightarrow 6\text{CO}_2 + 6\text{H}_2\text{O} + 36\text{ATP}$
- Glycolysis (cytoplasm) is common to BOTH aerobic and anaerobic respiration
- Krebs cycle is in the mitochondrial matrix; ETC is on the inner mitochondrial membrane